A Project Report On

Dictionary and Hybrid Attack on Login Services in VMware

Submitted By

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B.Sc. IT (IMS) Semester - VI

Internal Guide

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Jan – May 2026

Project Profile

Project Title	Dictionary and Hybrid Attack on Login Services in VMware
Project Type	Final Year Project — Cybersecurity Domain
Platform	Kali Linux (Attacker) + Red Hat Linux (Target)
Attack Targets	SSH (Port 22), FTP (Port 21), Web Login — DVWA (Port 80)
Attack Techniques	Dictionary Attack and Hybrid Attack
Wordlist Used	rockyou.txt (14 Million Passwords) + custom.txt
Dashboard	Custom Web Interface Built with Python Flask
Database	SQLite — Stores All Attack Logs and Sessions
Report	Auto-Generated PDF Report Using ReportLab Library

Introduction

Dictionary and Hybrid Attacks use automated tools and large password lists to attempt thousands of login combinations in a very short time. These attacks exploit common password patterns and human behavior to gain unauthorized access to accounts.

- Login services like SSH, FTP, and web applications are the most common entry points to any system and are widely used in real-world organizations and enterprises.
- Most users choose simple and predictable passwords such as names, dates, and common words — making these services easy targets for automated password attacks.
- This project focuses on executing dictionary and hybrid attacks on multiple login services and analyzing their success rate, speed, and impact in a controlled VMware lab environment.
- A custom web dashboard built using Python Flask is used to launch attacks, monitor live logs, visualize attempt speed through real-time charts, and generate downloadable PDF reports for each attack session.

Tools and Technologies Used

Attacker Side — Kali Linux

Tool	Description
Kali Linux	A Debian-based Linux OS specifically designed for penetration testing. It comes with 600+ pre-installed security tools. Used as the attacker machine to launch all dictionary and hybrid attacks.
Hydra	Fast network login cracker supporting over 50 protocols including SSH, FTP, HTTP, Telnet. Core tool used to perform both dictionary and hybrid attacks against all three login services.
Python + Flask	Python web framework used to build the custom attack monitoring dashboard with live log streaming, real-time charts, session history, and PDF report generation.
rockyou.txt	The most popular password wordlist containing 14 million real passwords leaked from real-world data breaches. Pre-installed in Kali Linux at /usr/share/wordlists/.
custom.txt	A custom wordlist created for this project with targeted base words. Used as input for the hybrid attack where Python automatically generates variations of each word.
SQLite	Lightweight embedded database that stores all attack session details, Hydra log lines, and results automatically in results.db without needing a separate database server.
ReportLab	Python library used to generate downloadable PDF reports for every attack session containing session details, target info, and the complete attack log.

Target Side — Red Hat Linux

Tool	Description
Red Hat Linux	Enterprise Linux OS — one of the most widely used server platforms in organizations worldwide. Configured as the victim machine running all three login services.
OpenSSH	Standard SSH server providing secure remote login on port 22. First login service target in this project.
vsftpd	Very Secure FTP Daemon — provides file transfer login on port 21. Second login service target in this project.
Apache + DVWA	Apache web server hosts DVWA — Damn Vulnerable Web Application — on port 80. DVWA provides a real intentionally insecure PHP login form as the third attack target.
MariaDB	Default MySQL replacement on Red Hat. Stores DVWA user database. Apache and PHP connect to MariaDB to verify login credentials when a user attempts to log in.
VMware	Virtualization platform running both VMs on one machine connected via host-only network — creating a completely isolated and safe lab environment.

Dictionary Attack

Definition

A dictionary attack is a method of attempting to gain unauthorized access to a system by systematically trying every word or phrase from a pre-compiled list of commonly used passwords against a login service until the correct one is found.

How It Works

- Hydra reads the wordlist file (rockyou.txt or custom.txt) line by line
- Tries each password against SSH, FTP, or Web login automatically
- Shows FAIL or SUCCESS for every single attempt in real time
- Stops immediately when the correct password is found

Hydra Command Used

```
hydra -l testuser -P wordlists/custom.txt ssh://192.168.126.140 -V -f -t 4
```

Command Flags Explained

Flag	Meaning
-l testuser	Username to attack
-P custom.txt	Password wordlist file to use
ssh://target	Service type and target IP address
-V	Show every attempt live in the terminal and dashboard
-f	Stop immediately when password is found
-t 4	Run 4 parallel threads for faster attack speed

Results in This Project

- SSH (port 22) — Password dcs@123 cracked in 12 attempts
- FTP (port 21) — Password dcs@123 cracked in 12 attempts
- DVWA Web Login — Password 'password' cracked in just 4 attempts

Hybrid Attack

Definition

A hybrid attack is an advanced extension of the dictionary attack that automatically generates hundreds of intelligent variations of each word by applying mutation rules such as adding numbers, appending symbols, and changing letter case — making it capable of cracking passwords that pure dictionary attacks would miss entirely.

How It Works

- Python code reads the base wordlist (custom.txt) automatically
- For every word it generates 10 variations using rules
- Saves the expanded list to logs/sessionid_hybrid.txt
- Hydra runs with this bigger expanded list against the target

Auto-Generated Variations Example

Base Word	Generated Variation	Why It Works
password	password123	Users add numbers at the end
admin	Admin	Users capitalize first letter
dcs	dcs@123	Users add symbols for security
redhat	REDHAT	Users use full uppercase
welcome	welcome2024	Users add current year

Python Code — Hybrid Wordlist Generator

```
for w in base_words:
    hybrid += [w, w+'123', w+'@123', w+'1',
               w+'!', w.capitalize(), w.upper(), w+'2024']
```

Results in This Project

- 10 base words from custom.txt generated 98 password variations automatically
- SSH hybrid attack cracked password dcs@123 in attempt number 84 out of 98
- FTP hybrid attack cracked password dcs@123 in attempt number 86 out of 98
- Hybrid attack found dcs@123 which is a variation not directly in the base wordlist

Dictionary Attack vs Hybrid Attack

Point	Dictionary Attack	Hybrid Attack
Definition	Tries passwords exactly as listed in wordlist	Generates and tries intelligent variations of each word
Wordlist	Pre-compiled wordlist used as-is	Wordlist + auto-generated mutations
Example	Tries only 'password'	Tries password, Password, password123, PASSWORD, password!
Attempts in demo	12-15 attempts	84-98 attempts
Limitation	Misses passwords with slight modifications	Takes more attempts due to more variations
Tool used	Hydra with wordlist directly	Python mutation + Hydra with expanded list
Best for	Exact common passwords	Modified common words with numbers or symbols

Key Takeaway

Dictionary attack is fast and works on simple passwords. Hybrid attack is more intelligent — it can crack passwords that users think are strong just because they added a number or symbol to a common word. In our project dcs@123 was cracked only by the hybrid attack because it is a variation not found directly in the wordlist.

Reason for Using Kali Linux and Red Hat Linux

Why Kali Linux — Attacker Machine

Kali Linux is the world's most popular penetration testing and ethical hacking operating system developed by Offensive Security. It comes with over 600 pre-installed cybersecurity tools and is widely used by security professionals worldwide.

- Hydra is pre-installed — the core tool used for all attacks in this project
- Professional terminal environment ideal for running and monitoring attacks
- Widely used in real-world penetration testing — makes project professionally relevant
- Host-only networking in VMware allows completely isolated and safe lab testing

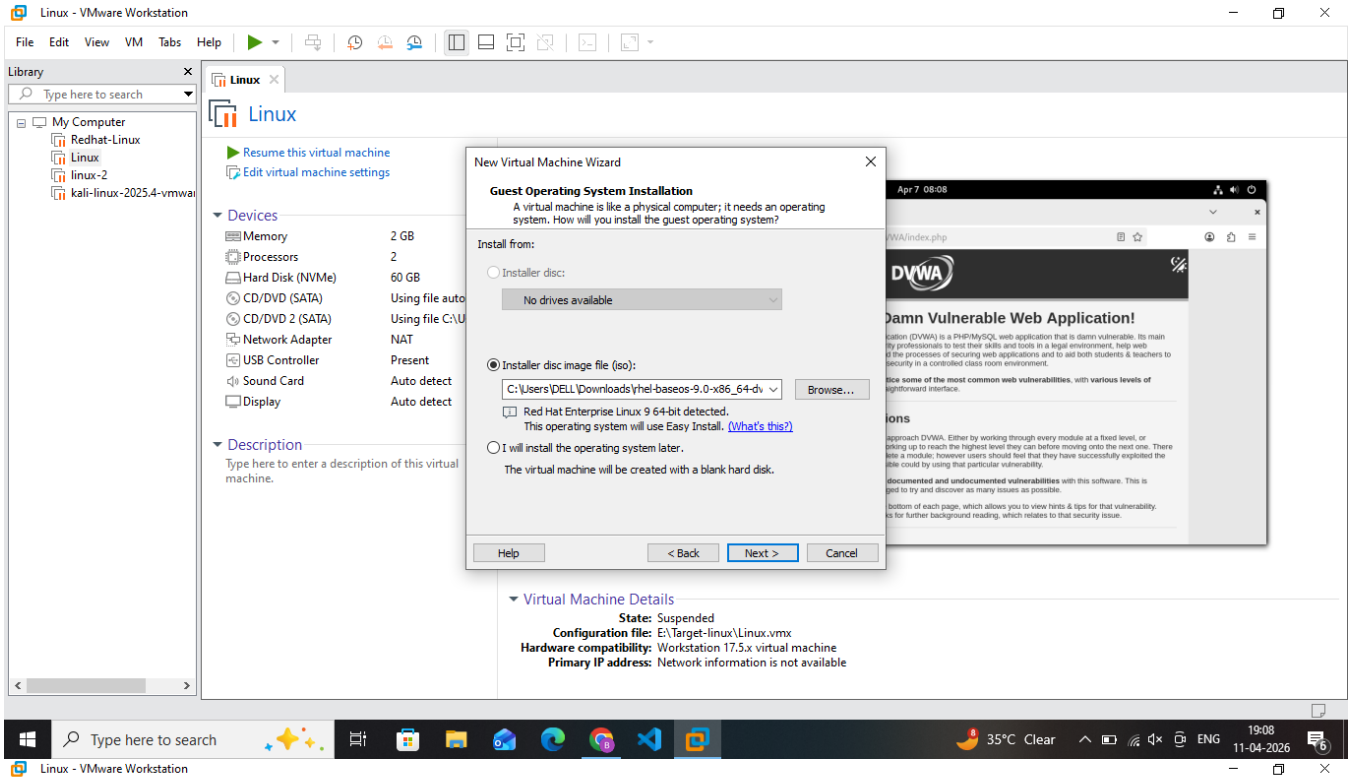
Why Red Hat Linux — Target Machine

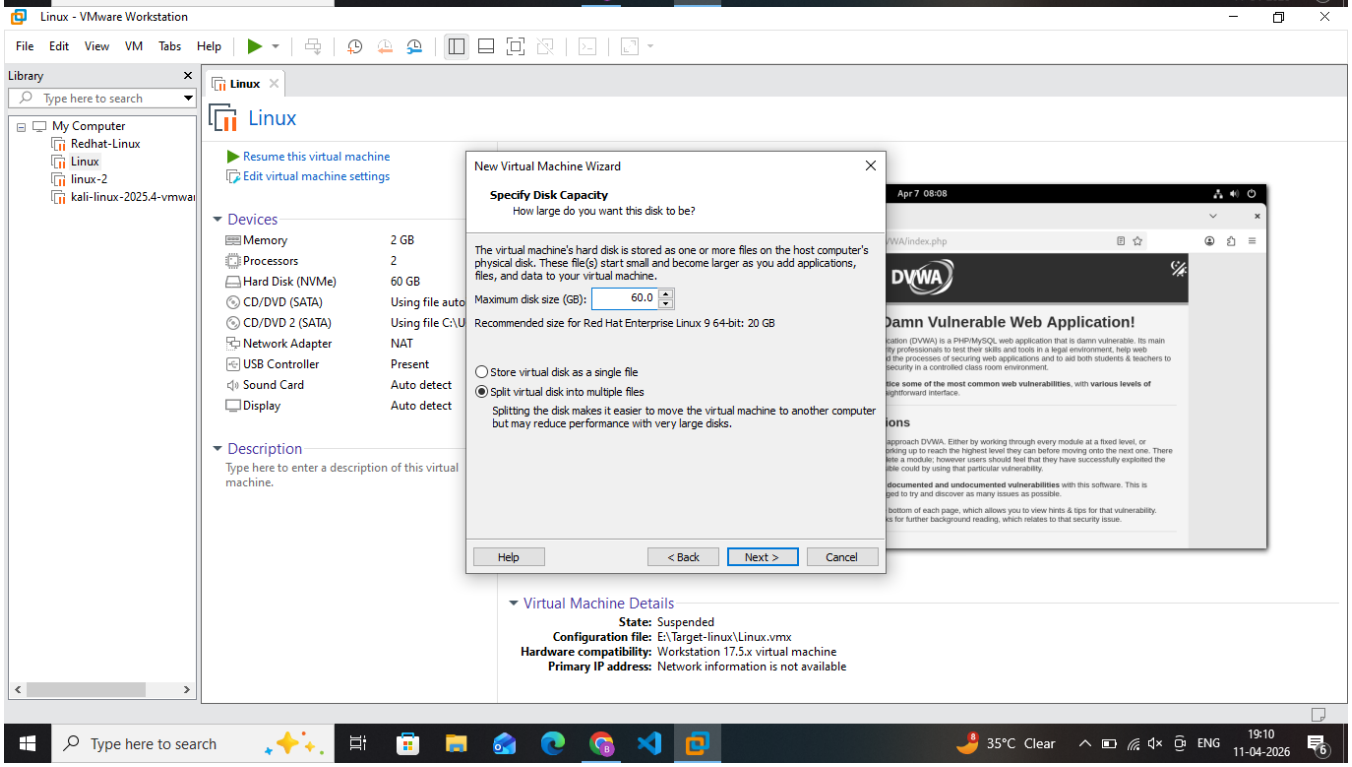
Red Hat Enterprise Linux is one of the most widely deployed Linux distributions in enterprise environments. Banks, government organizations, hospitals, and large companies use Red Hat to run their production servers.

- Represents a real enterprise server environment — makes demo realistic
- Runs OpenSSH, vsftpd, and Apache — real production-grade login services
- Uses SELinux — additional security layer that makes setup more technical and impressive
- Makes attack demonstration directly applicable to real-world cybersecurity scenarios

Why VMware Workstation

- Runs both VMs simultaneously on one physical laptop
- Host-only network creates completely isolated environment — no real systems affected
- Same approach used by professional penetration testers and cybersecurity labs
- Safe and legal environment to demonstrate attacks ethically


Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Redhat-Linux
Linux
linux-2
kali-linux-2025.4-vmwai
Linux
Resume this virtual machine
Edit virtual machine settings
Devices
Memory 2 GB
Processors 2
Hard Disk (NVMe) 60 GB
CD/DVD (SATA) Using file auto
CD/DVD 2 (SATA) Using file C:\U
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
C:\Users\DELL\Downloads\rhel-baseos-9.0-x86_64-dv
Browse...
Red Hat Enterprise Linux 9 64-bit detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel
Apr 7 08:08
DVWA
Damn Vulnerable Web Application!
Virtual Machine Details
State: Suspended
Configuration file: E:\Target-linux\Linux.vmx
Hardware compatibility: Workstation 17.5.x virtual machine
Primary IP address: Network information is not available
Type here to search
35°C Clear
ENG
19:08
11-04-2026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CD/DVD 2 (SATA) Using file C:\U
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 60.0
Recommended size for Red Hat Enterprise Linux 9 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel
Apr 7 08:08
DVWA
Damn Vulnerable Web Application!
Virtual Machine Details
State: Suspended
Configuration file: E:\Target-linux\Linux.vmx
Hardware compatibility: Workstation 17.5.x virtual machine
Primary IP address: Network information is not available
Type here to search
35°C Clear
ENG
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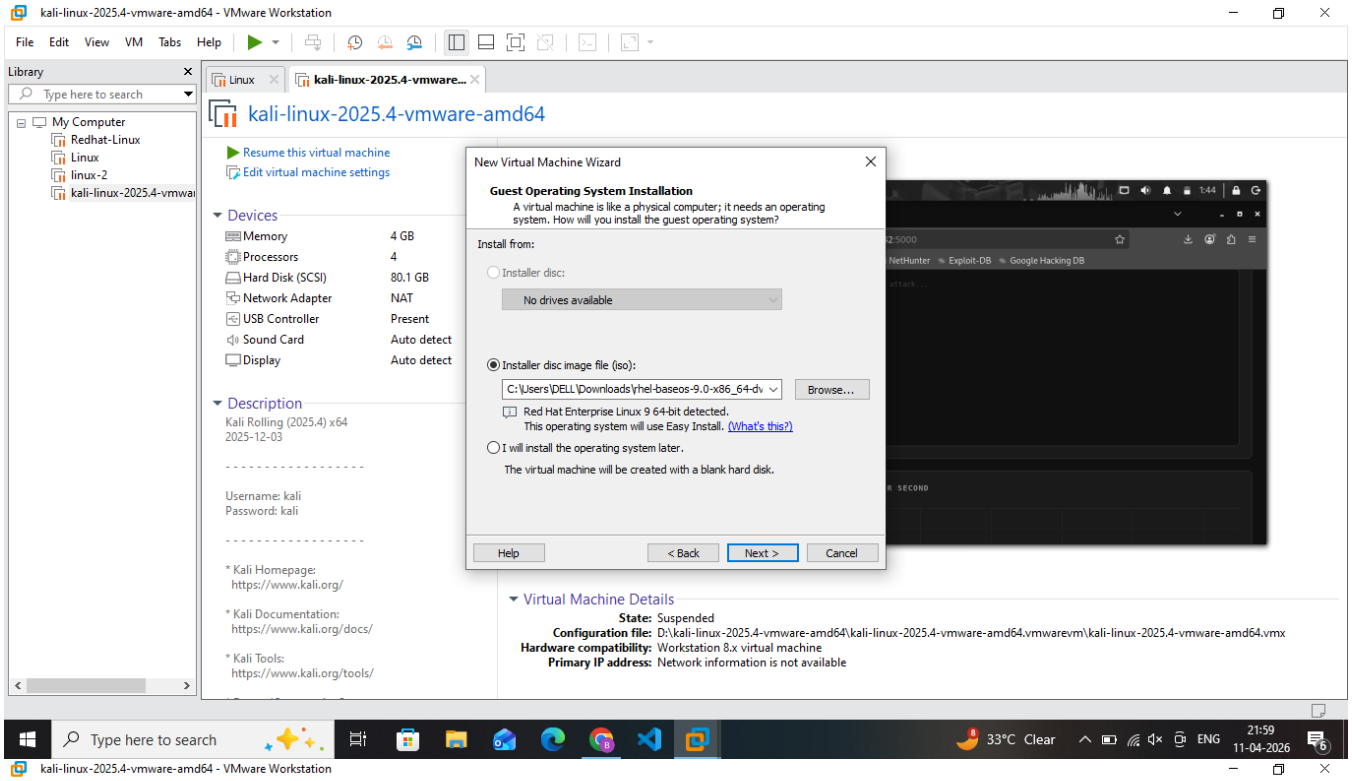
11-04-2026


kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Redhat-Linux
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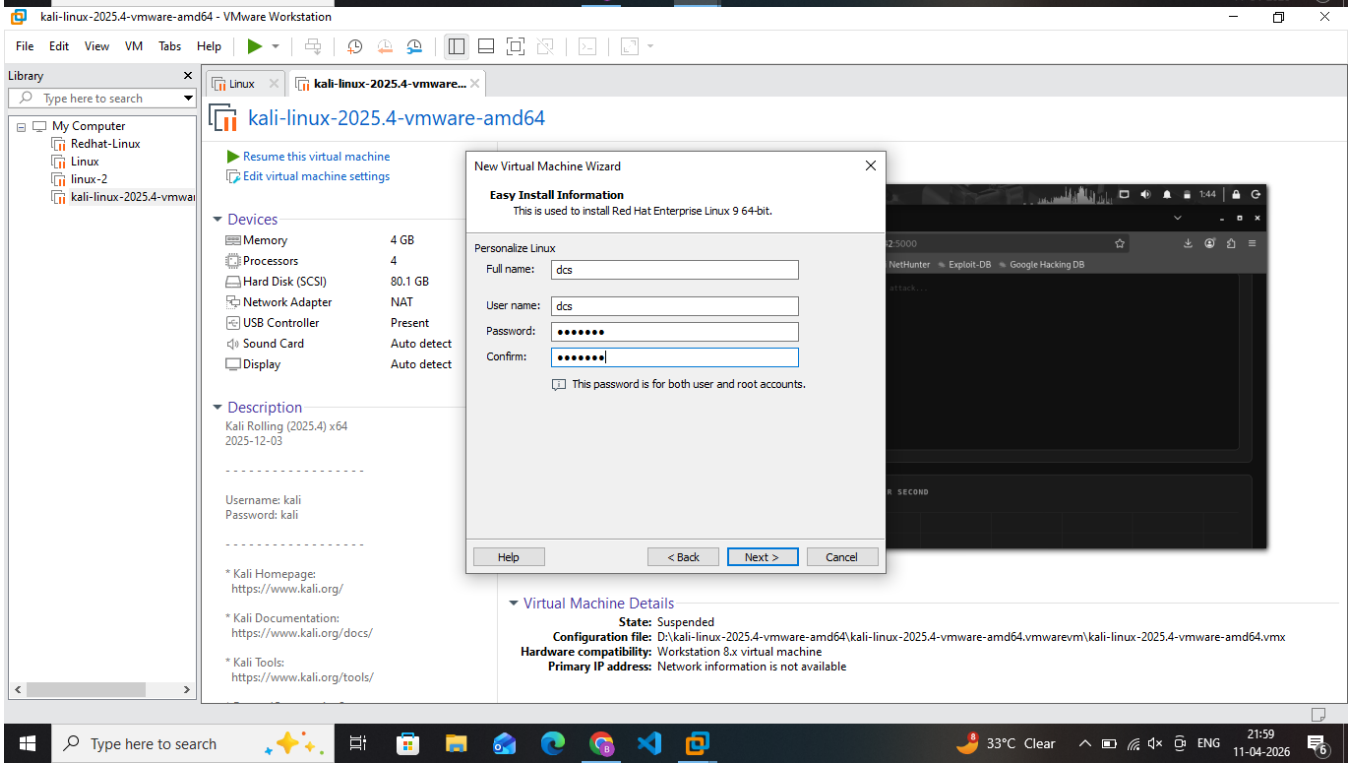
linux-2
kali-linux-2025.4-vmwai
Linux
kali-linux-2025.4-vmware...
kali-linux-2025.4-vmware-amd64
Resume this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 4
Hard Disk (SCSI) 80.1 GB
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Kali Rolling (2025.4) x64
2025-12-03
Username: kali
Password: kali
* Kali Homepage:
https://www.kali.org/
* Kali Documentation:
https://www.kali.org/docs/
* Kali Tools:
https://www.kali.org/tools/
New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
C:\Users\DELL\Downloads\rhel-baseos-9.0-x86_64-dv
Browse...
Red Hat Enterprise Linux 9 64-bit detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Suspended
Configuration file: D:\kali-linux-2025.4-vmware-amd64\kali-linux-2025.4-vmware-amd64.vmwarevm\kali-linux-2025.4-vmware-amd64.vmx
Hardware compatibility: Workstation 8.x virtual machine
Primary IP address: Network information is not available
Type here to search
33°C Clear
ENG
21:59
11-04-2026


kali-linux-2025.4-vmware-amd64 - VMware Workstation
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kali-linux-2025.4-vmware-amd64
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* Kali Documentation:
https://www.kali.org/docs/
* Kali Tools:
https://www.kali.org/tools/
New Virtual Machine Wizard
Easy Install Information
This is used to install Red Hat Enterprise Linux 9 64-bit.
Personalize Linux
Full name: dcs
User name: dcs
Password: •••••••
Confirm: •••••••
This password is for both user and root accounts.
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Suspended
Configuration file: D:\kali-linux-2025.4-vmware-amd64\kali-linux-2025.4-vmware-amd64.vmwarevm\kali-linux-2025.4-vmware-amd64.vmx
Hardware compatibility: Workstation 8.x virtual machine
Primary IP address: Network information is not available
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21:59
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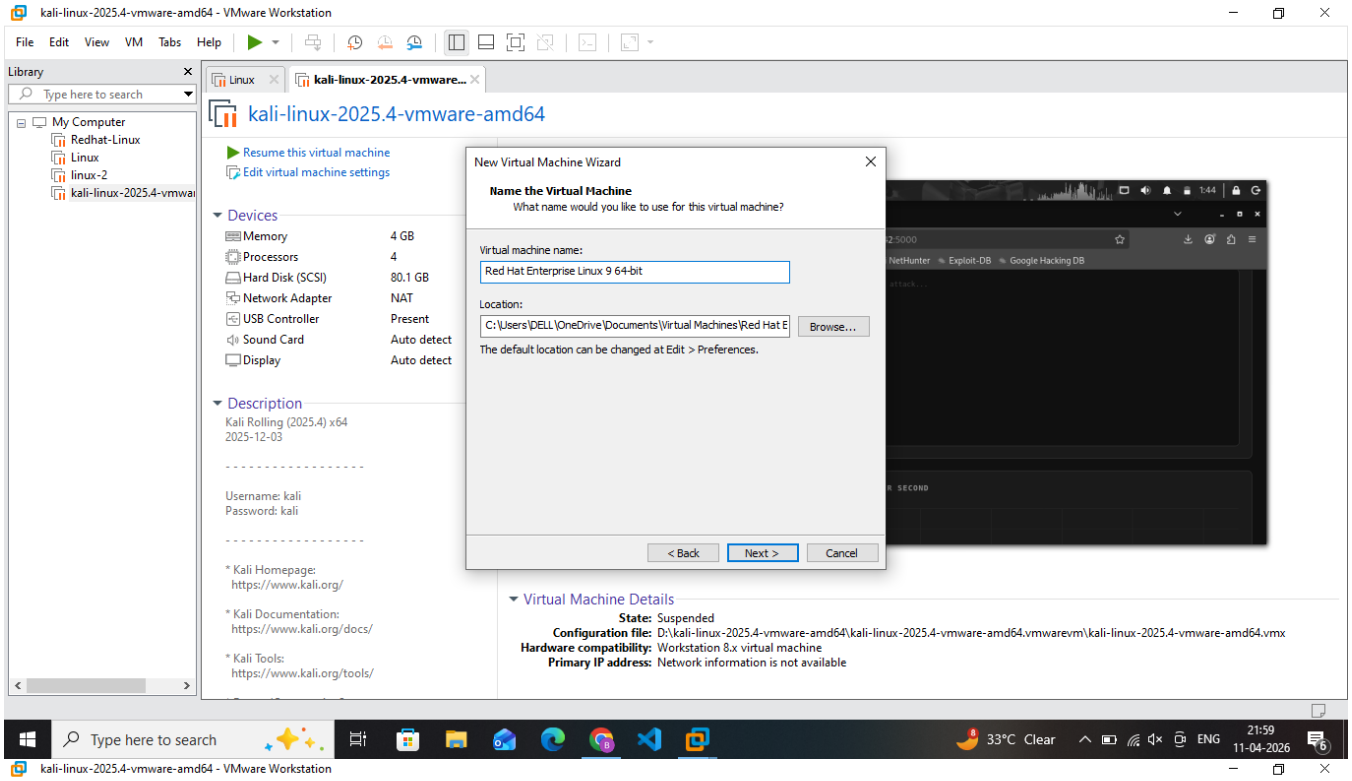
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Password: kali
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https://www.kali.org/
* Kali Documentation:
https://www.kali.org/docs/
* Kali Tools:
https://www.kali.org/tools/
New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Red Hat Enterprise Linux 9 64-bit
Location:
C:\Users\DELL\OneDrive\Documents\Virtual Machines\Red Hat E
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel
Virtual Machine Details
State: Suspended
Configuration file: D:\kali-linux-2025.4-vmware-amd64\kali-linux-2025.4-vmware-amd64.vmwarevm\kali-linux-2025.4-vmware-amd64.vmx
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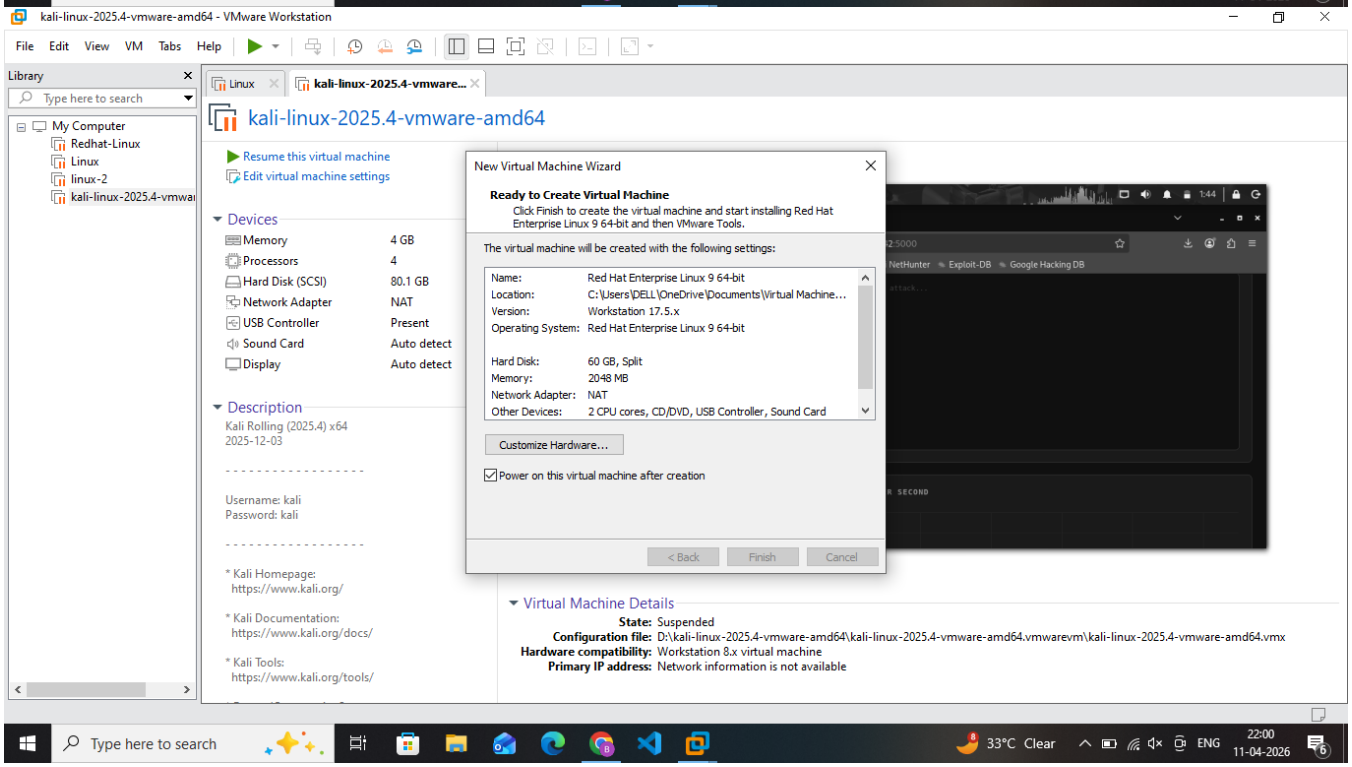
Type here to search
33°C Clear
ENG
21:59
11-04-2026


kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
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kali-linux-2025.4-vmware...
kali-linux-2025.4-vmware-amd64
Resume this virtual machine
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2025-12-03
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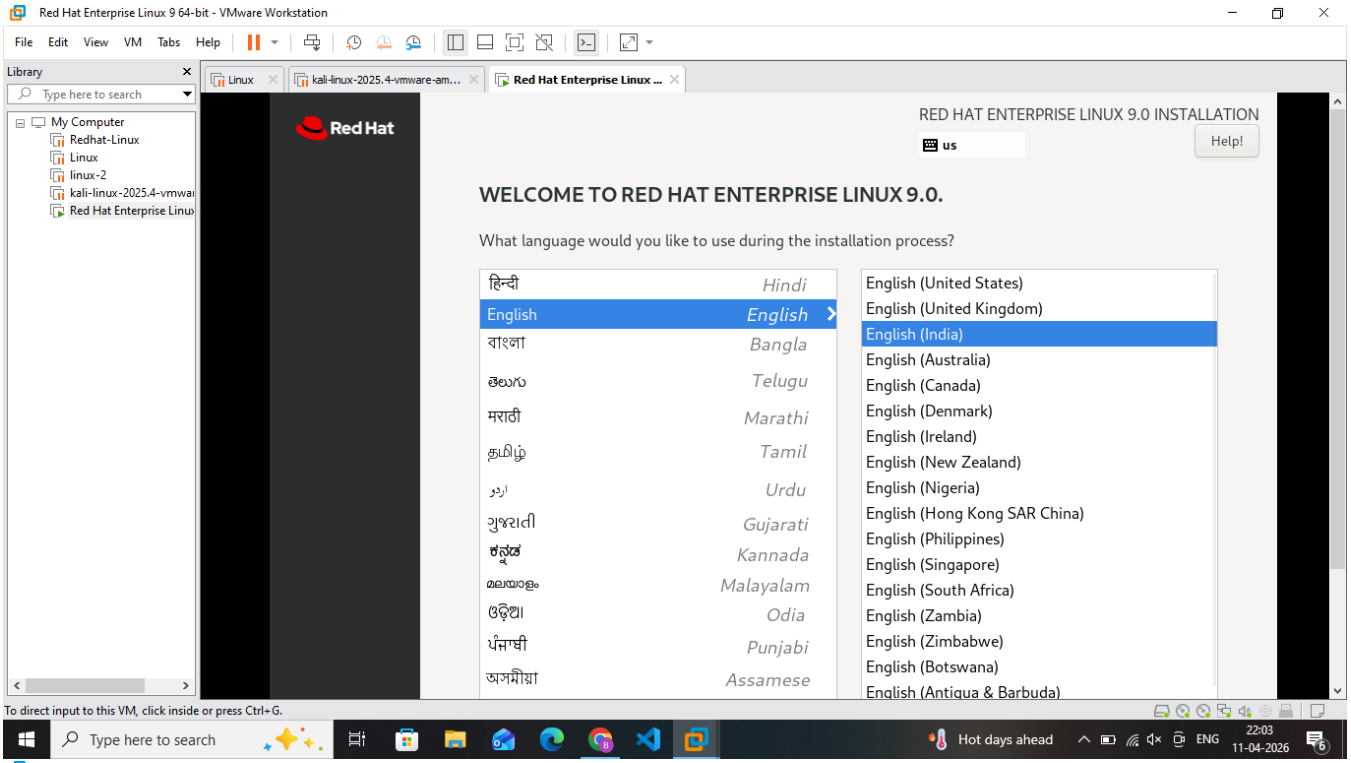
Password: kali
* Kali Homepage:
https://www.kali.org/
* Kali Documentation:
https://www.kali.org/docs/
* Kali Tools:
https://www.kali.org/tools/
New Virtual Machine Wizard
Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Red Hat Enterprise Linux 9 64-bit and then VMware Tools.
The virtual machine will be created with the following settings:
Name: Red Hat Enterprise Linux 9 64-bit
Location: C:\Users\DELL\OneDrive\Documents\Virtual Machine...
Version: Workstation 17.5.x
Operating System: Red Hat Enterprise Linux 9 64-bit
Hard Disk: 60 GB, Split
Memory: 2048 MB
Network Adapter: NAT
Other Devices: 2 CPU cores, CD/DVD, USB Controller, Sound Card
Customize Hardware...
Power on this virtual machine after creation
< Back
Finish
Cancel
Virtual Machine Details
State: Suspended
Configuration file: D:\kali-linux-2025.4-vmware-amd64\kali-linux-2025.4-vmware-amd64.vmwarevm\kali-linux-2025.4-vmware-amd64.vmx
Hardware compatibility: Workstation 8.x virtual machine
Primary IP address: Network information is not available
Type here to search
33°C Clear
ENG
22:00
11-04-2026


Red Hat Enterprise Linux 9 64-bit - VMware Workstation
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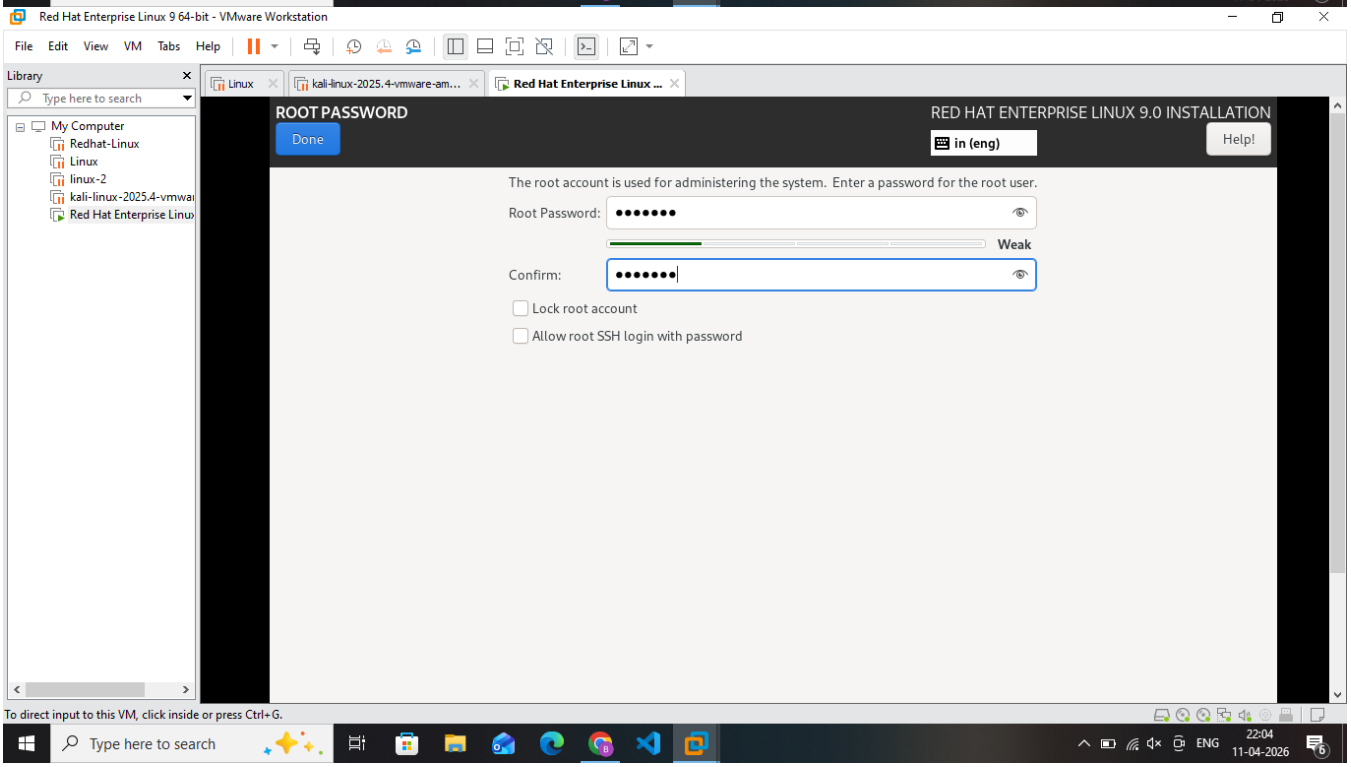
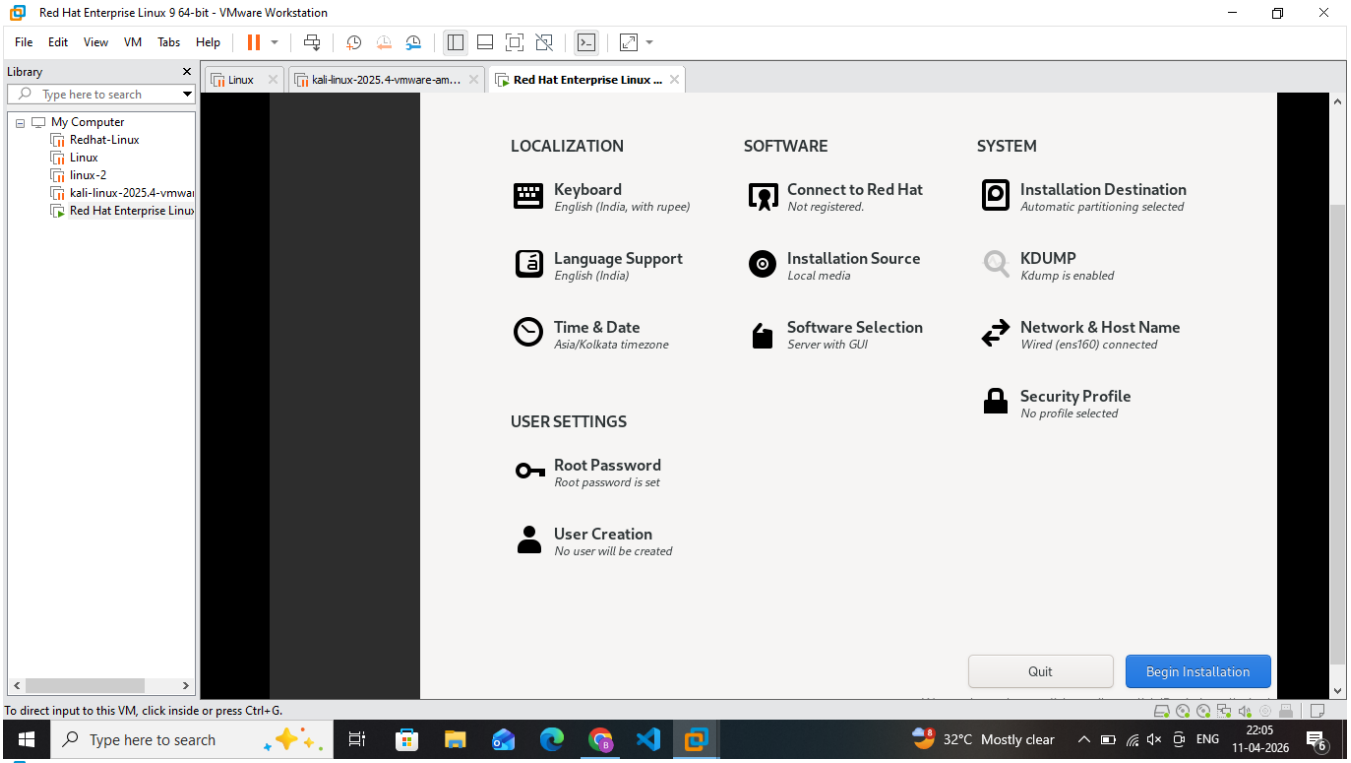
Red Hat Enterprise Linu
Linux
kali-linux-2025.4-vmware-am...
Red Hat Enterprise Linux ...
Red Hat
RED HAT ENTERPRISE LINUX 9.0 INSTALLATION
us
Help!
WELCOME TO RED HAT ENTERPRISE LINUX 9.0.
What language would you like to use during the installation process?
हिन्दी Hindi
English English
বাংলা Bangla
తెలుగు Telugu
मराठी Marathi
தமிழ் Tamil
اردو Urdu
ગુજરાતી Gujarati
ಕನ್ನಡ Kannada
മലയാളം Malayalam
ଓଡ଼ିଆ Odia
ਪੰਜਾਬੀ Punjabi
অসমীয়া Assamese
English (United States)
English (United Kingdom)
English (India)
English (Australia)
English (Canada)
English (Denmark)
English (Ireland)
English (New Zealand)
English (Nigeria)
English (Hong Kong SAR China)
English (Philippines)
English (Singapore)
English (South Africa)
English (Zambia)
English (Zimbabwe)
English (Botswana)
English (Antigua & Barbuda)
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
Hot days ahead
ENG
22:03
11-04-2026


Red Hat Enterprise Linux 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Redhat-Linux
Linux
linux-2
kali-linux-2025.4-vmwai
Red Hat Enterprise Linu
Linux
kali-linux-2025.4-vmware-am...
Red Hat Enterprise Linux ...
ROOT PASSWORD
Done
RED HAT ENTERPRISE LINUX 9.0 INSTALLATION
in (eng)
Help!
The root account is used for administering the system. Enter a password for the root user.
Root Password:
Weak
Confirm:
Lock root account
Allow root SSH login with password
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
ENG
22:04
11-04-2026


Red Hat Enterprise Linux 9 64-bit - VMware Workstation
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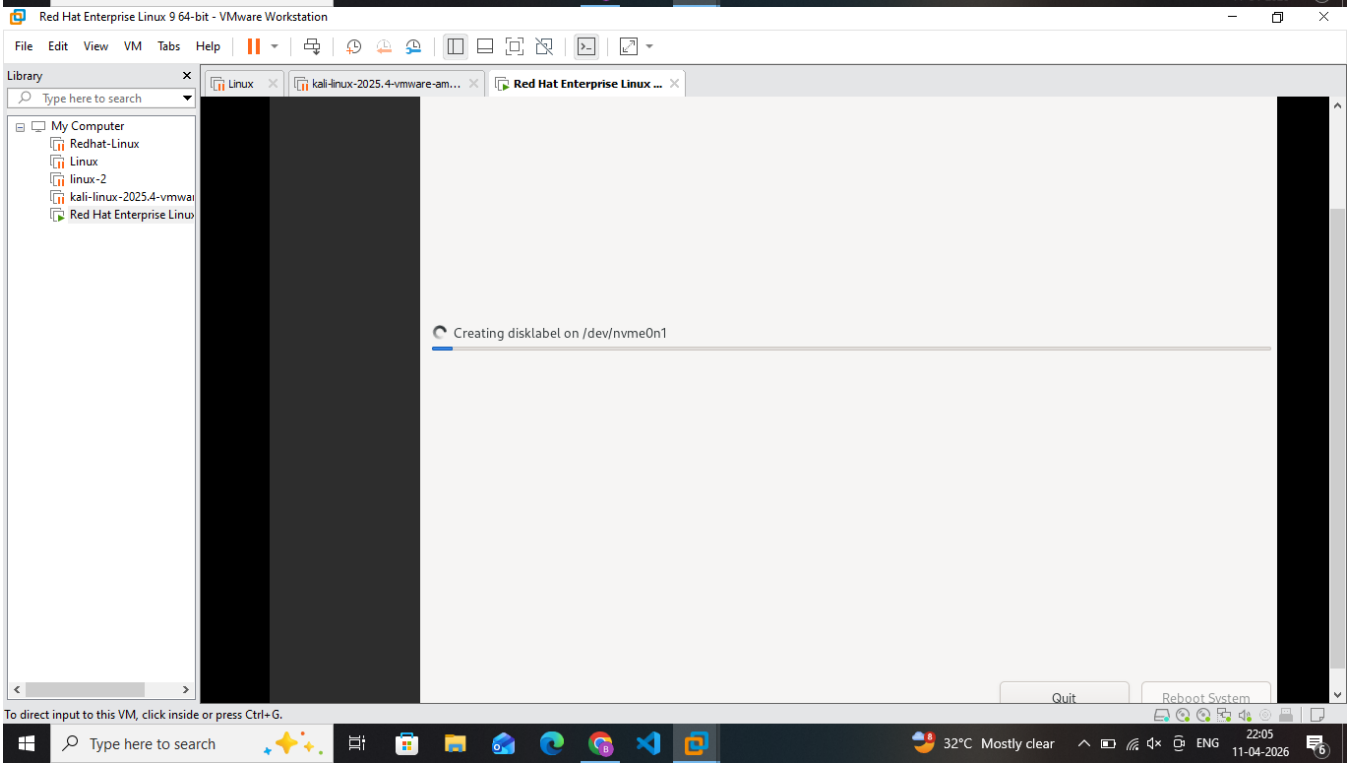
linux-2
kali-linux-2025.4-vmwai
Red Hat Enterprise Linu
Linux
kali-linux-2025.4-vmware-am...
Red Hat Enterprise Linux ...
LOCALIZATION
Keyboard
English (India, with rupee)
Language Support
English (India)
Time & Date
Asia/Kolkata timezone
SOFTWARE
Connect to Red Hat
Not registered.
Installation Source
Local media
Software Selection
Server with GUI
SYSTEM
Installation Destination
Automatic partitioning selected
KDUMP
Kdump is enabled
Network & Host Name
Wired (ens160) connected
Security Profile
No profile selected
USER SETTINGS
Root Password
Root password is set
User Creation
No user will be created
Quit
Begin Installation
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
32°C Mostly clear
ENG
22:05
11-04-2026


Red Hat Enterprise Linux 9 64-bit - VMware Workstation
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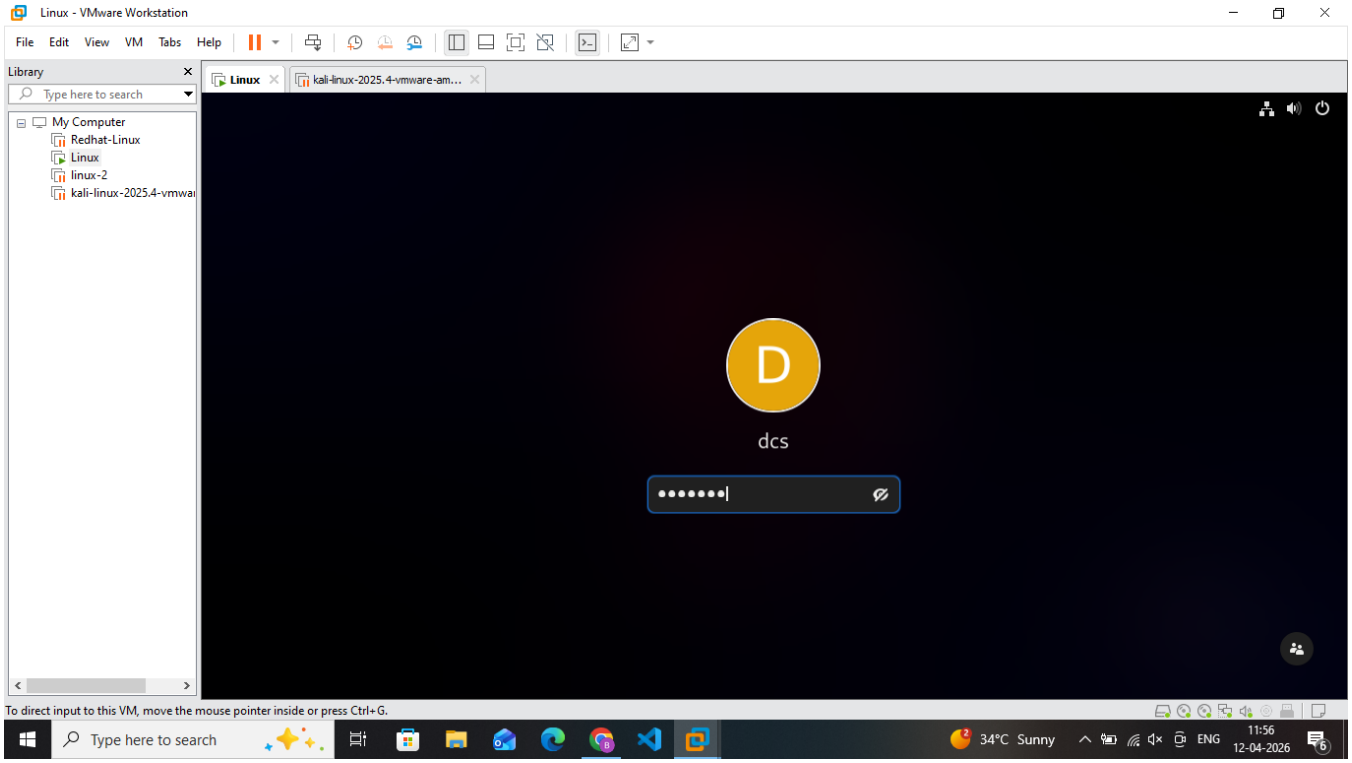
My Computer
Redhat-Linux
Linux
linux-2
kali-linux-2025.4-vmwai
Red Hat Enterprise Linu
Linux
kali-linux-2025.4-vmware-am...
Red Hat Enterprise Linux ...
Creating disklabel on /dev/nvme0n1
Quit
Reboot System
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
32°C Mostly clear
ENG
22:05
11-04-2026


Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Redhat-Linux
Linux
linux-2
kali-linux-2025.4-vmwai
Linux
kali-linux-2025.4-vmware-am...
D
dcs
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
34°C Sunny
ENG
11:56
12-04-2026

Conclusion

This project successfully demonstrated dictionary and hybrid attacks on three login services — SSH, FTP, and DVWA web login — in a controlled VMware lab environment.

- The dictionary attack directly tried passwords from the wordlist and cracked the login within seconds.
- The hybrid attack generated automatic variations of each word and cracked passwords that would be missed by a plain dictionary attack.
- A custom Python Flask dashboard was built to launch attacks, stream live logs, display real-time charts, save session history, and export PDF reports.
- The project proves that weak and predictable passwords can be cracked very easily using automated tools — highlighting the need for strong password policies in any real system.

Future Enhancement

1. **Cracking Encrypted Passwords (Hash Cracking)**

Currently the project attacks live login services directly. A major future enhancement is to extract password hashes from the target system such as the /etc/shadow file on Linux and crack them offline using Hashcat with GPU acceleration — demonstrating how attackers crack encrypted passwords without direct access to the login service.

2. **Multi-Factor Authentication (MFA) Testing**

Enhance the system to evaluate how dictionary and hybrid attacks perform when additional security layers such as Multi-Factor Authentication are implemented on the target — showing whether MFA successfully blocks automated attacks and providing a complete attack and defense perspective.

3. **AI-Based Password Generation**

Integrate machine learning models to generate smarter and more targeted password guesses based on user behavior patterns and previously leaked password datasets — instead of relying on traditional static wordlists — making hybrid attacks significantly more intelligent than rule-based mutations.

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